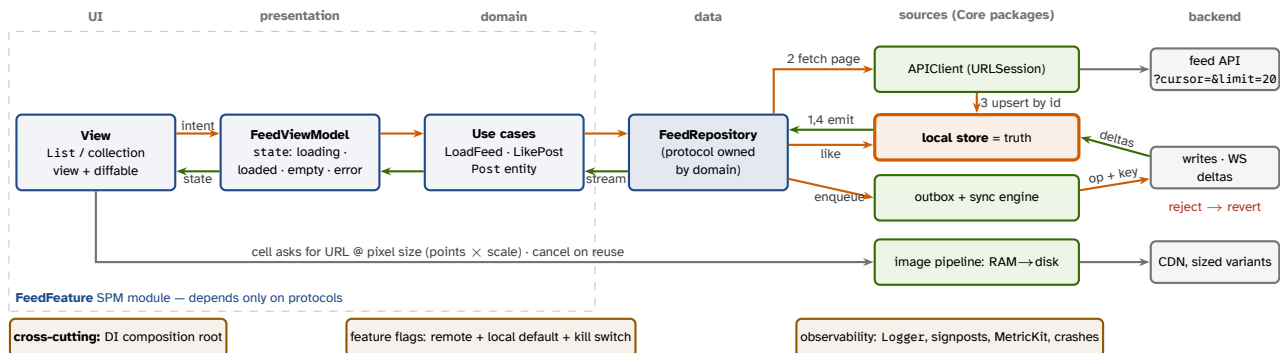


In one line: Order of decisions: **requirements** (functional, non-functional, scale) → **API + data model** → **client layers** → **data flow** (read + write) → the hard parts (sync, pagination, images) → **trade-offs**. On the client the **local store is the source of truth**; the network *syncs into* it.



The design, step by step

- **1 Requirements.** Functional: scroll, infinite load, pull-to-refresh, like, detail. Out of scope: upload, stories, DMs. Non-functional: cached feed *instantly* on launch, no hitches (16.7/8.3 ms), flaky network/offline, memory-safe images. Ranking + fan-out live on the backend; the client consumes a ranked list.
- **2 API + model.** GET /feed?cursor=<opaque>&limit=20 → {items, nextCursor}; items carry multi-resolution image URLs and viewerHasLiked. Normalize: entities by id in one store, the feed = ordered *id list* → one like updates every screen.
- **3 Layers.** View → ViewModel (UI state enum, no UIKit) → use cases → **Repository** (hides remote vs local) → data sources. Dependencies point inward; the domain owns the repository protocol (DIP) → fake repo in tests.
- **4 Data flow.** *Read:* observe local store (1) → fetch page (2) → upsert + dedup (3) → store emits (4). *Write:* optimistic store write → outbox → server → ack or **revert**.
- **Offline-first + sync.** Push dirty ops, pull *since cursor*. Retries need **idempotency** (client UUID key; “set” not “increment”). Backoff + jitter; per-item success. Deletes = **tombstones**. Conflicts — *name the policy*: server-wins, LWW (loses edits; clock skew), version/ETag + If-Match (412), field merge, CRDT. Triggers: foreground, NWPathMonitor, BGAppRefreshTask, silent push.
- **Pagination.** *Cursor/keyset:* stable under inserts, opaque, resumable. *Offset:* new posts at the top shift pages → duplicates and gaps. Prefetch the next page N rows before the end.
- **Images.** Memory: NSCache of *decoded, downsampled* images (cost limit, purged under pressure). Disk: size-bounded LRU of bytes. Downsample with ImageIO to points × scale, off main; coalesce duplicate URL requests; **cancel on reuse** and check the cell still shows the same id; prefetch via prefetchItemsAt.
- **Real-time.** WebSocket deltas for *visible* posts in the foreground; APNs when backgrounded. Server count *replaces*, never adds to, the optimistic one.
- **Platform.** SPM feature + Core modules (build time, boundaries). Flags: cached remote config, safe default. Token in **Keychain**; 401 → refresh + replay.

Example — repository: read local, sync in

```
actor FeedRepository: FeedRepositoryProtocol { // init omitted
    let api: APIClient; let store: PostStore; let outbox: Outbox
    private var cursor: String? // opaque, server-made
    func posts() -> AsyncStream<[Post]> { store.observeFeed() }
    func loadNext() async throws {
        let page = try await api.feed(after: cursor, limit: 20)
        try await store.upsert(page.items) // dedup by id -> emits
        cursor = page.nextCursor // nil = end of feed
    }
    func like(_ id: Post.ID) async {
        await store.setLiked(id, true) // optimistic, instant
        await outbox.enqueue(.like(id), key: UUID()) // retry-safe
    }
}
```

Trade-offs

Choice	What decides it
SwiftData · GRDB · files	queries + migrations · SQL control · feed cache only
SwiftUI List vs UIKit CV	velocity vs fine reuse / prefetch control
WebSocket · polling · APNs	live in foreground · fallback · app closed
Nuke / Kingfisher	use one — but justify each layer

Traps

- ! Network as source of truth → blank screen offline and on every cold launch.
- ! **Offset** pagination on a live feed; loading only at the bottom.
- ! Retrying a POST with no idempotency key → double like.
- ! Caching only bytes (decode per display); full-res decode on main.
- ! “It just syncs” — name the conflict policy and how you *detect* a conflict (version, not clock).
- ! **try!** decoding the page — one bad item kills the feed.

Remember

Requirements → **API** → **layers** → **flow** → **hard parts** → **trade-offs**. Local is truth · cursor · outbox + key · decoded RAM + disk LRU.